

Lecture Notes: Variational Quantum Algorithms

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Mostly due to [1] and [2].

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1 Variational Quantum Algorithms

Key point: A VQA is not a single algorithm. It is a design pattern for hybrid computation:

1. Choose a variational ansatz
2. Define a measurable objective, estimate that objective on quantum hardware.
3. Optimize the circuit parameters classically.

Fault-tolerant quantum computers would support deep circuits for algorithms such as phase estimation, Shor’s factoring algorithm, and Hamiltonian simulation. Historically variational quantum algorithms (VQAs) were motivated as follows: Current and near-term devices have limited qubit counts, restricted connectivity, finite coherence times, gate errors, and measurement errors. These constraints make deep coherent algorithms difficult to run before error correction becomes available [3].

1.1 Motivation (NISQ)

VQAs match near-term hardware in several ways:

- **Shallow circuits:** the ansatz can be chosen to respect the native gate set and connectivity of a device.
- **Hybrid error tolerance:** coherent calibration errors in parameterized gates may sometimes be absorbed by re-optimization.
- **Task-specific structure:** physical symmetries, locality, conservation laws, and graph structure can be built into the ansatz or cost.
- **Flexible outputs:** depending on the task, a VQA can output a state, an energy, a probability distribution, a bit string, a gate sequence, or a learned model.

The trade-off is that VQAs replace coherent circuit depth by repeated sampling and classical optimization. This moves the bottleneck from circuit depth to trainability, measurement cost, and noise robustness.

Key point. We want to trade off quantum resources, which are still limited, for classical resources. A VQA starts with a task, encodes it in a cost function $C(\theta)$, and then solves

$$\theta^* = \arg \min_{\theta} C(\theta). \tag{1}$$

The cost is estimated from measurement data produced by a parameterized quantum circuit. The estimated value, and sometimes its gradient, is passed to a classical optimizer. The optimizer proposes new parameters, and the loop repeats.

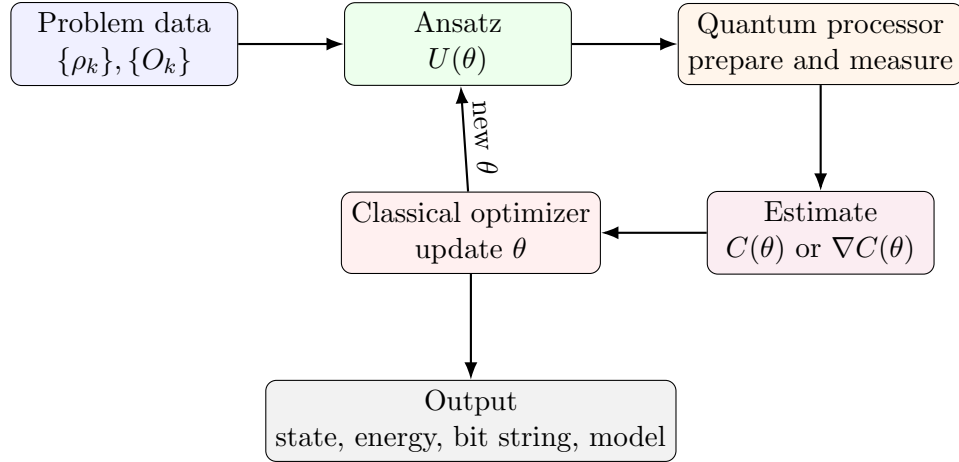


Figure 1: Schematic hybrid quantum-classical loop for a VQA.

1.2 Applications

Application class	Representative objective	Typical output
Ground states (Focus of this lecture)	Minimize $\langle \psi(\theta) H \psi(\theta) \rangle$	Energy estimate, state preparation
Quantum error correction	Penalize difference between states with error channels applied to them	Noise channel
Combinatorial optimization	Minimize or maximize encoded H_P	High-quality bit strings
Dynamics	Project time evolution onto $ \psi(\theta(t))\rangle$	Time-dependent observables
Compilation	Maximize fidelity between $U(\theta)$ and target V	Short native gate sequence
Machine learning	Minimize supervised or generative loss	Classifier, generator, feature map

1.3 Motivation (FT)

Since the main topic of this summer school is fault-tolerance, I will discuss where I believe VQAs could play a role in the FT regime.

- We are typically interested in eigenvalue problems and preparing ground states of physical systems.
- Given a Hamiltonian H , what is the ground state energy and corresponding eigenvector?
- QPE is a powerful, fault-tolerant algorithm that can achieve this.
- A powerful subroutine in quantum algorithms which forms the basis for Shor's algorithms and Quantum Phase Estimation is the Quantum Fourier Transform.

For an r -qubit register, let $N = 2^r$. The quantum Fourier transform QFT_N is defined by its action on computational basis states:

$$\text{QFT}_N |x\rangle = \frac{1}{2^{r/2}} \sum_{y=0}^{N-1} e^{2\pi i xy/N} |y\rangle. \quad (2)$$

Thus the QFT converts a computational-basis label x into a phase pattern over all computational basis states. In phase estimation, the inverse QFT performs the reverse task: it converts a phase pattern into a measurable binary string.

Example: two-qubit register. For two qubits, $N = 4$, and we write

$$|0\rangle = |00\rangle, \quad |1\rangle = |01\rangle, \quad |2\rangle = |10\rangle, \quad |3\rangle = |11\rangle.$$

The two-qubit QFT is

$$\text{QFT}_4 |x\rangle = \frac{1}{2} \sum_{y=0}^3 e^{2\pi i xy/4} |y\rangle. \quad (3)$$

For example,

$$\text{QFT}_4 |01\rangle = \frac{1}{2} (|00\rangle + i |01\rangle - |10\rangle - i |11\rangle). \quad (4)$$

So the input $|01\rangle$ is mapped to an equal superposition whose amplitudes have phases

$$1, \quad i, \quad -1, \quad -i.$$

The phase pattern is what the inverse QFT later decodes into a bit string in quantum phase estimation.

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix}.$$

Four qubit QFT:

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix}. \quad (5)$$

With this subroutine, we can now explore quantum phase estimation.

Quantum phase estimation estimates an eigenvalue of a unitary operator. Suppose

$$U |\phi_k\rangle = e^{2\pi i \varphi_k} |\phi_k\rangle, \quad \varphi_k \in [0, 1). \quad (6)$$

The algorithm uses an r -qubit phase register initialized in $|0\rangle^{\otimes r}$ and a system register initialized in an eigenstate $|\phi_k\rangle$. After applying Hadamards to the phase register, we have

$$\frac{1}{2^{r/2}} \sum_{\ell=0}^{2^r-1} |\ell\rangle |\phi_k\rangle. \quad (7)$$

Then controlled powers of U are applied:

$$|\ell\rangle |\phi_k\rangle \mapsto |\ell\rangle U^\ell |\phi_k\rangle. \quad (8)$$

Using the eigenvalue equation,

$$U^\ell |\phi_k\rangle = e^{2\pi i \ell \varphi_k} |\phi_k\rangle, \quad (9)$$

the joint state becomes

$$\frac{1}{2^{r/2}} \sum_{\ell=0}^{2^r-1} e^{2\pi i \ell \varphi_k} |\ell\rangle |\phi_k\rangle. \quad (10)$$

The phase register now contains the Fourier encoding of φ_k . Applying the inverse quantum Fourier transform gives

$$\text{QFT}^\dagger \frac{1}{2^{r/2}} \sum_{\ell=0}^{2^r-1} e^{2\pi i \ell \varphi_k} |\ell\rangle |\phi_k\rangle = |\tilde{\varphi}_k\rangle |\phi_k\rangle, \quad (11)$$

where $\tilde{\varphi}_k$ is an r -bit approximation to φ_k . Measuring the phase register therefore produces an estimate of the eigenphase.

For Hamiltonian problems, one often takes

$$U = e^{-iHt}, \quad (12)$$

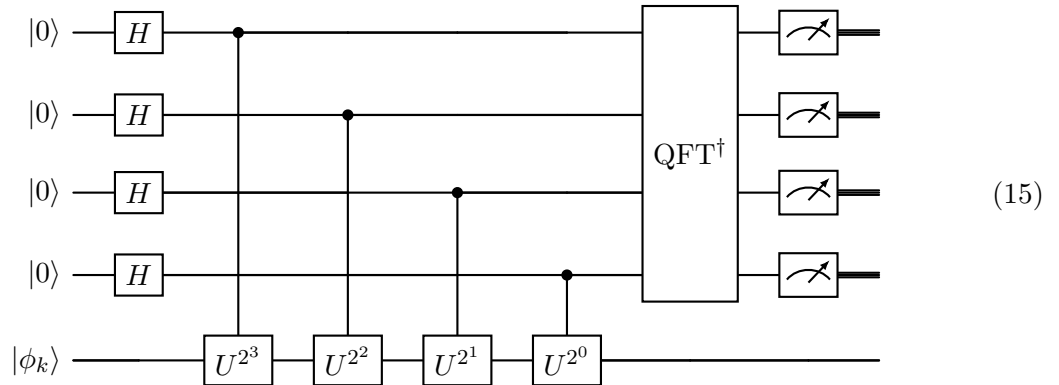
so that if

$$H |E_k\rangle = E_k |E_k\rangle, \quad (13)$$

then

$$U |E_k\rangle = e^{-iE_k t} |E_k\rangle. \quad (14)$$

Thus estimating the phase of U gives the energy E_k .



Question: What are the things we need?

Answer: We need oracle access to U , which has to be achieved via Hamiltonian simulation, will be discussed next lecture. We also are dealing with finite precision.

The above derivation assumes that the input state is an exact eigenstate. More generally, suppose the initial state is a superposition of energy eigenstates.

$$|\psi_0\rangle = \sum_k c_k |E_k\rangle. \quad (16)$$

By linearity, phase estimation produces

$$\sum_k c_k |\tilde{\varphi}_k\rangle |E_k\rangle. \quad (17)$$

After measuring the phase register, one obtains the energy E_k with probability approximately

$$|c_k|^2 = |\langle E_k | \psi_0 \rangle|^2. \quad (18)$$

Therefore, the probability of observing the ground-state energy E_0 is controlled by the initial ground-state overlap

$$p_0 = |\langle E_0 | \psi_0 \rangle|^2. \quad (19)$$

This overlap is crucial. If p_0 is large, then a small number of repetitions of phase estimation is likely to reveal the ground-state energy. If p_0 is exponentially small, then one needs roughly

$$O\left(\frac{1}{p_0}\right) \quad (20)$$

repetitions to see the ground state with constant probability. Thus quantum phase estimation is powerful when one can prepare an initial state with non-negligible overlap with the desired eigenstate, but it is not by itself a solution to the state-preparation problem.

Key point: We need finite overlap with the ground state. Mean field, prepare state with DMRG, perform VQE, perform QITE, perform Double Bracket Flow.

1.4 VQA Design

1. Choose a variational ansatz
2. Define a measurable objective, estimate that objective on quantum hardware.
3. Optimize the circuit parameters classically.

1.4.1 Variational Ansatz

I will denote a variational state by $|\psi(\theta)\rangle : \mathbb{R}^d \rightarrow \mathcal{H}$, which is normalized: $\langle\psi(\theta)|\psi(\theta)\rangle = 1$.

Key point: Note that $d = \text{poly}(n)$. The variational manifold is denoted by $\mathcal{M}$.

The word “variational” comes from replacing an intractable search over all states or operations by a search over a restricted, parameterized family.

Key point: This is also done classically with for example Tensor Networks and Neural Quantum States.

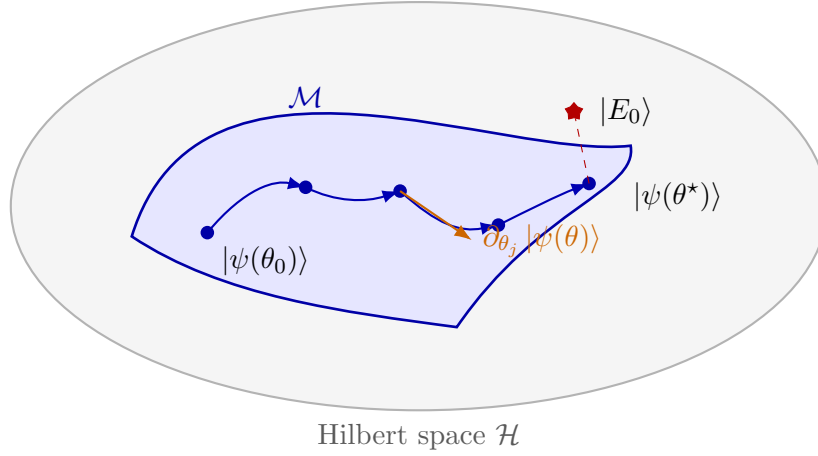


Figure 2: A variational ansatz defines a low-dimensional manifold $\mathcal{M}$ of accessible quantum states inside the full Hilbert space. The optimizer moves along this manifold by changing the circuit parameters θ . If the true ground state $|E_0\rangle$ is not contained in $\mathcal{M}$, the remaining distance represents ansatz error.

The ansatz is the parameterized family of states or operations being searched. Often

$$|\psi(\theta)\rangle = U(\theta) |\psi_0\rangle, \quad (21)$$

where

$$U(\theta) = U_L(\theta_L) \cdots U_2(\theta_2)U_1(\theta_1). \quad (22)$$

Each layer may be built from parameterized rotations, entangling gates, and fixed gates chosen to match the hardware.

Example 1 *Problem-agnostic ansatzes.* *Hardware-efficient ansatzes use native gates and connectivity to keep depth small. They are easy to implement but can be hard to train if they become too expressive or random.*

Example 2 *Problem-inspired ansatzes.* *These use domain knowledge. Examples include the unitary coupled-cluster ansatz in quantum chemistry, QAOA-style alternating ansatzes for optimization, Hamiltonian variational ansatzes for many-body systems, and symmetry-preserving circuits.*

Example 3 *Adaptive ansatzes.* *Variable-structure algorithms such as ADAPT-VQE grow the circuit during optimization by adding gates that appear most useful. The goal is to reduce unnecessary depth while retaining enough expressibility. We will get to these later!*

1.4.2 Objective functions

The cost function must encode the answer to the task. A general form is

$$C(\theta) = f(\{\rho_k\}, \{O_k\}, U(\theta)), \quad (23)$$

where $\{\rho_k\}$ are input states or training data, $\{O_k\}$ are observables, and $U(\theta)$ is the parameterized circuit. A common measurable structure is

$$C(\theta) = \sum_k f_k \left(\text{Tr} \left[O_k U(\theta) \rho_k U^\dagger(\theta) \right] \right). \quad (24)$$

A useful VQA cost should be:

- **Faithful:** its global minimum corresponds to the desired solution;
- **Measurable:** it can be estimated from a feasible number of quantum measurements;
- **Trainable:** the optimizer can find good parameters without exponential resources.

Key point: $C(\theta)$ is typically non-convex, which means there is no straightforward, provably convergent algorithm to find its minimum.

We will focus on the ground state problem, since this captures a wide class of problems. This is called VQE: variational quantum eigensolver.

Example 4 *For a Hamiltonian H , for example, the Rayleigh-Ritz variational principle says*

$$E_0 \leq \langle \psi(\theta) | H | \psi(\theta) \rangle \quad (25)$$

for any normalized trial state $|\psi(\theta)\rangle$. Minimizing the expectation value over θ gives an upper bound on the ground-state energy. This is the basis of the variational quantum eigensolver (VQE).

Key point: We will focus on the variational energy optimization, which is known as VQE.

We will denote Hamiltonians, which are Hermitian operators $H \in L(\mathcal{H})$ as a linear combination of $M = \text{poly}(n)$ terms

$$H = \sum_{m=1}^M O_m \quad (26)$$

where the O_m are Hermitian. Typical operators are the Pauli's,

$$\mathcal{P}_1 := \{X, Y, Z, I\}$$

and their tensor product

$$\mathcal{P}_n := \{a_1 \otimes \cdots \otimes a_n : a_i \in \mathcal{P}_1\}$$

which form a basis of $|\mathcal{P}_n| = 4^n$ of the Hermitian matrices. Note that the Paulis are idempotent: $P_i^2 = I$, $P_i \in \mathcal{P}_n$.

Suppose the Hamiltonian admits a decomposition into measurable observables, for example Pauli strings,

$$H = \sum_{m=1}^M h_m O_m. \quad (27)$$

Then, by linearity of expectation,

$$E(\theta) := \langle \psi(\theta) | H | \psi(\theta) \rangle = \sum_{m=1}^M h_m \langle \psi(\theta) | O_m | \psi(\theta) \rangle. \quad (28)$$

Thus the energy can be estimated by estimating each expectation value $\langle \psi(\theta) | O_m | \psi(\theta) \rangle$ separately.

For a Pauli observable O_m , each measurement outcome is a random variable $X_s \in \{-1, +1\}$ with mean

$$\mathbb{E}[X_s] = \langle \psi(\theta) | O_m | \psi(\theta) \rangle. \quad (29)$$

Using N_s shots, define the empirical estimator

$$\hat{O}_m = \frac{1}{N_s} \sum_{s=1}^{N_s} X_s. \quad (30)$$

Hoeffding's inequality gives

$$\Pr \left[\left| \hat{O}_m - \langle \psi(\theta) | O_m | \psi(\theta) \rangle \right| \geq \epsilon \right] \leq 2 \exp \left(-\frac{N_s \epsilon^2}{2} \right). \quad (31)$$

Therefore, to estimate one Pauli expectation value to additive error ϵ with failure probability at most δ , it is enough to take

$$N_s \geq \frac{2}{\epsilon^2} \log \left(\frac{2}{\delta} \right). \quad (32)$$

Hence,

$$N_s = O \left(\frac{1}{\epsilon^2} \log \frac{1}{\delta} \right), \quad (33)$$

or simply

$$N_s = O \left(\frac{1}{\epsilon^2} \right) \quad (34)$$

when suppressing logarithmic dependence on the failure probability.

Thus, provided the number of terms M grows at most polynomially with system size, the energy can be estimated efficiently to inverse-polynomial precision by estimating the individual terms in the Hamiltonian decomposition.

Key point. Estimating observables from QC is $\text{poly}(n)$

Example 5 For two quantum states $|\phi(\theta)\rangle$ and $|\psi\rangle$, we have the fidelity

$$\mathcal{F}(\psi, \phi) = |\langle \psi | \phi \rangle|^2 \leq 1 \quad (35)$$

Hadamard test for state fidelity. Suppose two states are prepared by circuits

$$U_\psi |0\rangle = |\psi\rangle, \quad U_\phi |0\rangle = |\phi\rangle. \quad (36)$$

The fidelity between the two pure states is

$$F(\psi, \phi) = |\langle \phi | \psi \rangle|^2. \quad (37)$$

Define

$$W = U_\phi^\dagger U_\psi. \quad (38)$$

Then

$$\langle 0 | W | 0 \rangle = \langle 0 | U_\phi^\dagger U_\psi | 0 \rangle = \langle \phi | \psi \rangle. \quad (39)$$

Thus estimating the overlap $\langle \phi | \psi \rangle$ is enough to estimate the fidelity.

The **Hadamard test** estimates the real part of $\langle 0 | W | 0 \rangle$. Start from

$$|0\rangle |0\rangle. \quad (40)$$

Apply a Hadamard gate to the ancilla:

$$\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |0\rangle. \quad (41)$$

Then apply a controlled- W , controlled on the ancilla:

$$\frac{1}{\sqrt{2}} (|0\rangle |0\rangle + |1\rangle W |0\rangle). \quad (42)$$

Finally apply another Hadamard to the ancilla. The state becomes

$$\frac{1}{2} |0\rangle (I + W) |0\rangle + \frac{1}{2} |1\rangle (I - W) |0\rangle. \quad (43)$$

Therefore

$$p(0) = \frac{1}{4} \langle 0 | (I + W^\dagger)(I + W) | 0 \rangle \quad (44)$$

$$= \frac{1}{2} + \frac{1}{2} \text{Re} \langle 0 | W | 0 \rangle. \quad (45)$$

Hence

$$\text{Re} \langle \phi | \psi \rangle = \text{Re} \langle 0 | W | 0 \rangle = 2p(0) - 1. \quad (46)$$

To estimate the imaginary part, one performs the same test but measures the ancilla in the Y -basis, giving

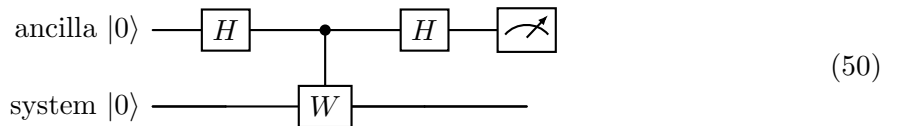
$$\text{Im} \langle \phi | \psi \rangle = \text{Im} \langle 0 | W | 0 \rangle. \quad (47)$$

Combining the two estimates gives

$$F(\psi, \phi) = |\langle \phi | \psi \rangle|^2 = (\text{Re} \langle \phi | \psi \rangle)^2 + (\text{Im} \langle \phi | \psi \rangle)^2. \quad (48)$$

If the overlap is known to be real, then a single Hadamard test is sufficient:

$$F(\psi, \phi) = (2p(0) - 1)^2. \quad (49)$$



1.4.3 Optimizing the cost function

Original [4, 5]. Final theory [6]. See also [7] for multivariate extensions.

Many VQA optimizers use gradients. A central practical observation is that, for many common parametrized quantum gates, these gradients can be obtained from ordinary circuit evaluations rather than by differentiating the circuit internally. This is the content of the *parameter-shift rule*.

Consider a cost function of the form

$$C(\theta) = \langle 0 | U^\dagger(\theta) O U(\theta) | 0 \rangle, \quad (51)$$

where O is an observable and $U(\theta)$ is a parametrized circuit. To derive the rule, isolate a single parameter θ_j and write the circuit as

$$U(\theta) = U_L U_j(\theta_j) U_R, \quad (52)$$

where U_L and U_R contain all gates before and after the parametrized gate, respectively. Suppose that

$$U_j(\theta_j) = e^{-i\theta_j P_j/2}, \quad (53)$$

where P_j is a Pauli operator, so that

$$P_j^2 = I. \quad (54)$$

Absorbing the surrounding gates into a state and an effective observable, define

$$|\psi\rangle = U_R |0\rangle, \quad \bar{O} = U_L^\dagger O U_L. \quad (55)$$

Then the dependence of the cost on the single parameter θ_j is

$$C(\theta_j) = \langle \psi | U_j^\dagger(\theta_j) \bar{O} U_j(\theta_j) | \psi \rangle. \quad (56)$$

Using $P_j^2 = I$, the exponential can be evaluated exactly:

$$U_j(\theta_j) = \cos\left(\frac{\theta_j}{2}\right) I - i \sin\left(\frac{\theta_j}{2}\right) P_j. \quad (57)$$

The derivative is then

$$\partial_{\theta_j} U_j(\theta_j) = \frac{-i}{2} P_j U_j(\theta_j) \quad (58)$$

from which we get

$$\partial_{\theta_j} C(\theta_j) = \frac{-i}{2} \langle \psi | \left[P_j, U_j^\dagger(\theta_j) \bar{O} U_j(\theta_j) \right] | \psi \rangle. \quad (59)$$

After some trigonometry

$$\partial_{\theta_j} C(\theta_j) = \frac{1}{2} \left[C\left(\theta_j + \frac{\pi}{2}\right) - C\left(\theta_j - \frac{\pi}{2}\right) \right]. \quad (60)$$

Key point This is not a finite-difference approximation. A finite-difference estimator would approximate the derivative by evaluating the cost at nearby parameter values and taking the limit

as the shift goes to zero. By contrast, Eq. (60) is an exact analytic identity for gates of the form

$$e^{-i\theta_j P_j/2} \quad (61)$$

with Pauli generator P_j . The only statistical error comes from estimating the shifted costs with finitely many measurement shots. Thus, the gradient estimator inherits shot noise, but it does not incur discretization error from the shift itself.

Key point. This cost is $O(d)$, so linear in the number of parameters, compare this to $O(1)$ in automatic differentiation.

The classical optimizer sees a noisy, non-convex objective. Common choices include:

- **gradient-based methods:** stochastic gradient descent, Adam, quantum natural gradient, imaginary-time-inspired updates;
- **gradient-free methods:** COBYLA, Nelder-Mead, SPSA, Bayesian optimization, evolutionary strategies;
- **problem-aware methods:** layerwise training, warm starts, parameter interpolation, local updates, and adaptive shot allocation.

The optimizer should be judged not only by iteration count, but by the number of quantum circuit evaluations, the number of measurements per evaluation, and its robustness to noise.

1.5 Technical issues

1.5.1 Measurement efficiency

Estimating $C(\theta)$ can require many quantum measurements. In VQE, for instance,

$$H = \sum_m O_m \quad (62)$$

so the energy estimate is built from many Pauli-string expectation values. For electronic-structure Hamiltonians, the number of terms can scale poorly with the number of orbitals.

Example 6 (Back-of-the-envelope chemistry cost) *In electronic-structure VQE, a fermion-to-qubit mapping often produces many Pauli terms. If all terms are measured separately and uniformly, the total measurement cost can be very large. For chemical accuracy,*

$$\epsilon \approx 10^{-3} \text{ Ha}, \quad (63)$$

the basic Monte Carlo scaling already gives a factor $1/\epsilon^2 \approx 10^6$. An example Hamiltonian is

$$H = \sum_{p,q} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{p,q,r,s} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s$$

For about 60 orbitals, there are $M \sim 10^5$ Pauli terms, a naive estimate can easily reach

$$N_{\text{total}} \sim 10^{11} \quad (64)$$

shots per accurately estimated energy. Grouping commuting terms, optimized shot allocation, low-rank decompositions, and other measurement reductions are therefore essential in practice.

Several methods reduce measurement overhead:

- group mutually commuting Pauli strings and measure them together;
- allocate more shots to high-variance or large-coefficient terms;
- use classical shadows or other randomized measurement schemes;
- use surrogate models or neural-network tomography, while tracking possible bias.

Measurement cost is often the practical bottleneck: an algorithm with shallow circuits may still be expensive if each optimizer step requires many circuit repetitions.

1.5.2 Noise and accuracy

Noise affects VQAs in several distinct ways:

- it can slow training by flattening the cost landscape;
- it can shift the optimum of the noisy cost away from the optimum of the ideal cost;
- it can corrupt the final estimated value even if training succeeds.

1.5.3 Barren plateaus

Original [8] and final theory[9]

The parameter-shift rule gives an exact expression for the gradient, but it does not guarantee that the gradient is large enough to estimate efficiently. The barren plateau problem is the statement that, for sufficiently random ansatzes, the gradient is exponentially concentrated around zero [8, 9].

Consider again a cost function

$$C(\theta) = \langle 0 | U^\dagger(\theta) O U(\theta) | 0 \rangle. \quad (65)$$

Isolate one parameter,

$$U(\theta) = U_L U_j(\theta_j) U_R, \quad U_j(\theta_j) = e^{-i\theta_j P_j/2}, \quad (66)$$

and define

$$|\psi\rangle = U_R |0\rangle, \quad \bar{O} = U_L^\dagger O U_L. \quad (67)$$

Then

$$C(\theta_j) = \langle \psi | U_j^\dagger(\theta_j) \bar{O} U_j(\theta_j) | \psi \rangle. \quad (68)$$

Differentiating gives

$$\partial_{\theta_j} C = \frac{i}{2} \langle \psi | \left[P_j, U_j^\dagger(\theta_j) \bar{O} U_j(\theta_j) \right] | \psi \rangle. \quad (69)$$

Equivalently, by absorbing $U_j(\theta_j)$ into the state, define

$$\rho_j = U_j(\theta_j) |\psi\rangle \langle \psi| U_j^\dagger(\theta_j), \quad (70)$$

so that

$$\partial_{\theta_j} C = \frac{i}{2} \text{Tr} [\rho_j [P_j, \bar{O}]]. \quad (71)$$

Now assume that the circuit blocks U_R and U_L are sufficiently deep and unstructured that they behave approximately as independent Haar-random unitaries, or as **unitary 2-designs**. Then ρ_j behaves like a random pure state and $\bar{O} = U_L^\dagger O U_L$ behaves like a random rotation of the observable.

First, the mean gradient vanishes. Since a Haar-random state satisfies

$$\mathbb{E}[\rho_j] = \frac{I}{d}, \quad d = 2^n, \quad (72)$$

we obtain

$$\mathbb{E}[\partial_{\theta_j} C] = \frac{i}{2} \text{Tr} \left[\frac{I}{d} [P_j, \bar{O}] \right] \quad (73)$$

$$= \frac{i}{2d} \text{Tr} [[P_j, \bar{O}]] \quad (74)$$

$$= 0, \quad (75)$$

because the trace of a commutator is zero.

The key point is the **variance**. Let

$$A = [P_j, \bar{O}]. \quad (76)$$

Then

$$\partial_{\theta_j} C = \frac{i}{2} \text{Tr}(\rho_j A). \quad (77)$$

For a Haar-random pure state, by **Weingarten calculus**,

$$\mathbb{E}[\rho_j \otimes \rho_j] = \frac{I + \text{SWAP}}{d(d+1)}, \quad (78)$$

Therefore,

$$\mathbb{E}_{\rho_j} [(\partial_{\theta_j} C)^2] = -\frac{1}{4} \mathbb{E}_{\rho_j} [\text{Tr}(\rho_j A)^2] \quad (79)$$

$$= -\frac{1}{4} \mathbb{E}_{\rho_j} [\text{Tr}(\rho_j A \otimes \rho_j A)] \quad (80)$$

$$= -\frac{1}{4d(d+1)} [\text{Tr}(A)^2 + \text{Tr}(A^2)]. \quad (81)$$

Since A is a commutator, $\text{Tr}(A) = 0$, so

$$\mathbb{E}_{\rho_j} [(\partial_{\theta_j} C)^2] = -\frac{1}{4d(d+1)} \text{Tr}(A^2). \quad (82)$$

Averaging also over the random observable $\bar{O} = U_L^\dagger O U_L$ gives the standard Haar-random barren-plateau scaling

$$\text{Var}(\partial_{\theta_j} C) = O\left(\frac{\text{Tr}(O^2)}{d^2}\right). \quad (83)$$

Since $d = 2^n$, this becomes

$$\text{Var}(\partial_{\theta_j} C) = O\left(\frac{\text{Tr}(O^2)}{4^n}\right). \quad (84)$$

For a typical global observable with $\text{Tr}(O^2) = O(2^n \text{poly}(n))$, this gives

$$\text{Var}(\partial_{\theta_j} C) = O\left(\frac{\text{poly}(n)}{2^n}\right). \quad (85)$$

Thus the gradient satisfies

$$\mathbb{E}[\partial_{\theta_j} C] = 0, \quad \text{Var}(\partial_{\theta_j} C) = O\left(\frac{\text{poly}(n)}{2^n}\right). \quad (86)$$

Therefore the typical gradient magnitude scales as

$$|\partial_{\theta_j} C| \sim O\left(\frac{\text{poly}(n)}{2^{n/2}}\right). \quad (87)$$

The cost landscape is then exponentially flat in the number of qubits. Since the gradient signal is exponentially small, resolving it with shot-based measurements requires exponentially many samples.

This is the barren plateau problem: for deep, highly expressive, random ansatzes, the gradients vanish not just on average, but with exponentially small variance.

1.5.4 Dynamical Lie algebras and trainability

A useful way to refine the barren plateau discussion is to replace the vague notion of “expressibility” by a precise algebraic object: the dynamical Lie algebra (DLA) generated by the circuit. Consider a parametrized circuit of the form

$$U(\theta) = \prod_{\ell=1}^L e^{-i\theta_\ell H_\ell}, \quad (88)$$

where the Hermitian generators H_ℓ are drawn from some generator set $\mathcal{G}$. The corresponding DLA is

$$\mathfrak{g} = \langle i\mathcal{G} \rangle_{\text{Lie}}, \quad (89)$$

namely the real vector space spanned by all nested commutators of the skew-Hermitian generators iH_ℓ . Equivalently,

$$\mathfrak{g} = \text{span}_{\mathbb{R}} \{iH_a, [iH_a, iH_b], [iH_a, [iH_b, iH_c]], \dots\}. \quad (90)$$

The reachable unitaries of a sufficiently expressive circuit lie in the Lie group

$$G = e^{\mathfrak{g}}. \quad (91)$$

Thus $\mathfrak{g}$ captures the subgroup of $SU(2^n)$ explored by the ansatz.

Example 7 an Abelian $\mathfrak{u}(1)$ DLA. Consider the two-site generator set

$$\mathcal{A} = \{XY\}. \quad (92)$$

Since there is only one generator, no new directions are produced by commutation:

$$[iXY, iXY] = 0. \quad (93)$$

Therefore

$$\langle iXY \rangle_{\text{Lie}} \cong \mathfrak{u}(1). \quad (94)$$

This is the simplest possible DLA: it generates a one-parameter subgroup.

Example 8 A two-local $\mathfrak{su}(2)$ DLA. A non-Abelian example is

$$\mathcal{A} = \{XX, XZ\}. \quad (95)$$

The commutator generates a third Pauli direction:

$$[iXX, iXZ] = 2i IY. \quad (96)$$

The three directions close under commutation:

$$[iXX, iXZ] = 2i IY, \quad (97)$$

$$[iXZ, iIY] = 2i XX, \quad (98)$$

$$[iIY, iXX] = 2i XZ. \quad (99)$$

Thus

$$\langle iXX, XZ \rangle_{\text{Lie}} \cong \mathfrak{su}(2). \quad (100)$$

This is a useful example because the generators are two-local, but their Lie closure is the familiar three-dimensional $\mathfrak{su}(2)$ algebra.

Example 9 Gaussian fermions. The set

$$\mathcal{A}_{\text{FF}} = \{XX, ZI, IZ\}. \quad (101)$$

generates the Lie algebra

$$\langle XX, ZI, IZ \rangle_{\text{Lie}} = \mathfrak{so}(4). \quad (102)$$

For an n -site chain, this family grows to

$$\mathfrak{a}_{14}(n) \cong \mathfrak{so}(2n), \quad (103)$$

which is the standard Lie algebra associated with quadratic fermionic Gaussian dynamics.

If the circuit is deep enough to behave like a unitary design over $G = e^{\mathfrak{g}}$, then the variance of the cost can be expressed in terms of the DLA. In the simplest case where $\mathfrak{g}$ is simple, one obtains schematically

$$\text{Var}_{\theta}[C(\theta)] = \frac{\mathcal{P}_{\mathfrak{g}}(\rho)\mathcal{P}_{\mathfrak{g}}(O)}{\dim(\mathfrak{g})}. \quad (104)$$

Equation (104) makes precise the connection between expressibility and barren plateaus. If $\dim(\mathfrak{g})$ grows exponentially with the number of qubits, then the variance of the cost, and similarly the variance of its gradients, can become exponentially small. In that case the landscape is exponentially concentrated, and resolving a descent direction requires exponentially many measurements. On the other hand, if $\dim(\mathfrak{g})$ grows only polynomially, then the size of the reachable group alone does not force a barren plateau. Barren plateaus may still occur because of the state or observable, but they are no longer caused simply by the ansatz being too expressive.

This motivates classifying variational ansatzes by their DLAs. Instead of asking whether a circuit is “expressive” in an informal sense, we ask:

$$\text{What is } \mathfrak{g} = \langle i\mathcal{G} \rangle_{\text{Lie}}, \quad \text{and how does } \dim(\mathfrak{g}) \text{ scale with } n? \quad (105)$$

In [10], we answer this question for a broad family of physically motivated generator sets: the Pauli terms appearing in 2-local spin systems on one-dimensional and fully connected topologies. The main message is that the resulting DLAs do not exhibit arbitrary scaling. Instead, for these 2-local spin-system generators, the DLA dimensions fall into three classes:

$$\dim(\mathfrak{g}) = O(n), \quad O(n^2), \quad \text{or} \quad O(4^n). \quad (106)$$

Therefore, within this family, the DLA immediately predicts the expected trainability regime:

- If $\dim(\mathfrak{g}) = O(4^n)$, the ansatz explores an exponentially large subgroup and one expects barren-plateau behavior from expressibility.
- If $\dim(\mathfrak{g}) = O(n)$ or $O(n^2)$, the ansatz explores only a polynomial-dimensional subgroup, so expressibility alone does not imply an exponentially small gradient variance.

This gives a practical geometric interpretation of circuit design. A generator pool does not merely specify gates; it specifies a Lie algebra $\mathfrak{g}$, and hence a reachable Lie group $e^{\mathfrak{g}}$. The dimension and structure of this algebra determine whether the circuit behaves like a highly expressive random ansatz or like a restricted, structured subgroup. The classification of spin-system DLAs therefore provides a map from physically motivated Hamiltonian terms to expected optimization behavior in variational quantum algorithms.

2 Double Bracket flows.

See [11], [12] and [13, 14]. Two fantastic references are [15] and [16]

2.1 Quantum circuits as paths on the unitary group

A quantum circuit on n qubits implements a unitary matrix. Since global phases do not affect quantum states, we may regard the circuit as an element of the special unitary group

$$SU(p) = \left\{ U \in \mathbb{C}^{p \times p} : U^\dagger U = I, \det(U) = 1 \right\}.$$

with $p = 2^n$. In the usual variational setting, one first chooses a parametrized map

$$\theta \in \mathbb{R}^m \mapsto U(\theta) \in SU(p),$$

and then optimizes over the coordinates θ . Geometrically, this means that the ansatz selects a restricted subset of the full unitary group. The resulting circuit can also be viewed as a discrete path on $SU(p)$:

$$I \longrightarrow U_1 \longrightarrow U_2 U_1 \longrightarrow \cdots \longrightarrow U_L \cdots U_2 U_1.$$

Each gate moves the circuit a small distance along the unitary group.

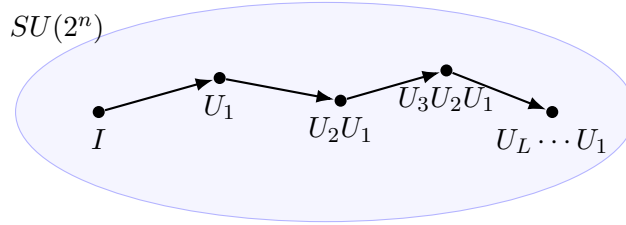


Figure 3: A circuit as a discrete path through the unitary group.

Key point. This motivates an alternative perspective is to optimize directly over $SU(p)$. This removes the need to choose coordinates θ at the outset. Instead, one asks for the steepest descent direction of the cost function on the manifold $SU(p)$ itself.

2.2 From Euclidean gradient descent to Riemannian gradient flows

An archetypal example of a gradient-based method in quantum computing is the Variational Quantum Eigensolver. Let

$$C(\theta) = \text{Tr} \left[H U(\theta) \rho_0 U(\theta)^\dagger \right],$$

The goal of VQE is to solve

$$\min_{\theta} C(\theta).$$

The usual gradient-descent update is

$$\theta_{k+1} = \theta_k - \eta \nabla_{\theta} C(\theta_k),$$

where $\eta > 0$ is a step size. This is a discretization of the continuous-time negative-gradient flow

$$\dot{\theta} = -\nabla_{\theta} C(\theta).$$

Why does the negative gradient decrease the cost? Let $\theta(t)$ be a curve in parameter space. By the chain rule,

$$\frac{d}{dt}C(\theta(t)) = \nabla_{\theta}C(\theta(t)) \cdot \dot{\theta}(t).$$

Or if we set $\theta(0) = x$ and $\dot{\theta}(0) = v$, we have that the **differential**

$$dC(x)(v) = \nabla_{\theta}C(x) \cdot v$$

If we choose

$$\dot{\theta}(t) = -\nabla_{\theta}C(\theta(t)),$$

then

$$\frac{d}{dt}C(\theta(t)) = -\|\nabla_{\theta}C(\theta(t))\|^2 \leq 0.$$

Thus the function value decreases monotonically along the continuous-time negative-gradient flow.

Geometrically, the gradient is perpendicular to the **level sets** of C . Suppose $\gamma(t)$ is a curve satisfying

$$C(\gamma(t)) = \text{constant}.$$

Then

$$0 = \frac{d}{dt}C(\gamma(t)) = \nabla_{\theta}C(\gamma(t)) \cdot \dot{\gamma}(t).$$

Therefore the gradient is orthogonal to every tangent direction $\dot{\gamma}(t)$ along the level set. Moving in the direction $-\nabla_{\theta}C$ gives the locally steepest decrease in the Euclidean metric.

The usual VQE optimization takes place in the coordinate space $\mathbb{R}^m$. However, the physically relevant object is the unitary $U(\theta)$. This motivates a coordinate-free question:

Question: How do we generalize these ideas to manifolds? Can we optimize directly over the unitary group instead of over a chosen circuit parameterization?

To answer this, we replace Euclidean gradient descent by Riemannian gradient descent on the unitary group.

2.3 Gradient flows on the unitary group

Let H and B be fixed Hermitian $p \times p$ matrices. We consider the objective

$$C(U) = \text{Tr} \left[HUBU^{\dagger} \right], \quad U \in \text{SU}(p).$$

The group $\text{SU}(p)$ is not a vector space, so ordinary Euclidean gradient descent does not directly apply. Instead, we need a notion of gradient that lives on the manifold $\text{SU}(p)$.

A Riemannian gradient requires two ingredients:

1. a tangent space at every point $U \in \text{SU}(p)$, describing the allowed infinitesimal directions of motion;

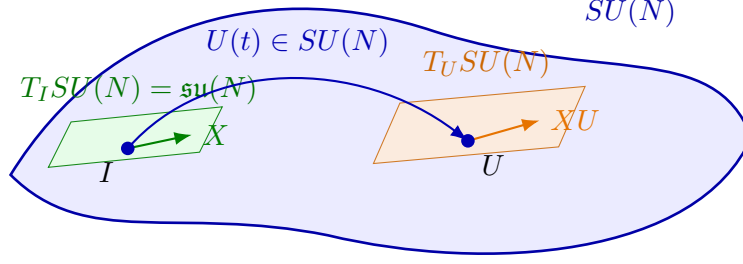


Figure 4: The special unitary group $SU(N)$ is a smooth manifold. Its tangent space at the identity is the Lie algebra $\mathfrak{su}(N)$, consisting of traceless skew-Hermitian matrices. Tangent vectors at a general point $U \in SU(N)$ are obtained by translating Lie-algebra elements from the identity to U , giving $T_U SU(N) = \{XU : X \in \mathfrak{su}(N)\}$.

2. an inner product on tangent vectors, called a Riemannian metric, which allows us to define steepest descent.

Tangent vectors on $SU(p)$. The tangent space at the identity is the Lie algebra

$$\mathfrak{su}(p) = \left\{ \Omega \in \mathbb{C}^{p \times p} : \Omega^\dagger = -\Omega, \text{Tr}(\Omega) = 0 \right\}.$$

Thus $\mathfrak{su}(p)$ consists of traceless skew-Hermitian matrices.

To see this, let $U(t)$ be a smooth curve in $SU(p)$ satisfying

$$U(0) = I, \quad \dot{U}(0) = \Omega.$$

Since $U(t)$ is unitary,

$$U(t)^\dagger U(t) = I.$$

Differentiating and evaluating at $t = 0$ gives

$$\Omega^\dagger + \Omega = 0,$$

so Ω is skew-Hermitian.

Since $U(t) \in SU(p)$, we also have

$$\det(U(t)) = 1.$$

Using

$$\frac{d}{dt} \det(U(t)) = \det(U(t)) \text{Tr} \left[U(t)^{-1} \dot{U}(t) \right],$$

and evaluating at $t = 0$, we obtain

$$\text{Tr}(\Omega) = 0.$$

Therefore

$$T_I SU(p) = \mathfrak{su}(p).$$

At a general point $U \in \text{SU}(p)$, tangent vectors are obtained by translating Lie-algebra elements from the identity to U :

$$T_U \text{SU}(p) = \{\Omega U : \Omega \in \mathfrak{su}(p)\}.$$

Thus any infinitesimal motion on the unitary group can be written as

$$\dot{U} = \Omega U, \quad \Omega \in \mathfrak{su}(p).$$

Now that we have tangent vectors, we can start thinking about gradients. In order to talk about gradients, we need notions of distances and angles.

The Riemannian gradient is defined using a metric. For a function on Euclidean space, the gradient is the vector satisfying

$$dC(x)(v) = \nabla C(x) \cdot v$$

for every direction v . On a manifold, the same idea applies, but the dot product is replaced by a Riemannian metric.

Let

$$XU, YU \in T_U \text{SU}(p), \quad X, Y \in \mathfrak{su}(p).$$

We choose the left-invariant Hilbert–Schmidt metric

$$g_U(A, B) = \text{Re Tr} \left[A^\dagger B \right].$$

Since $U^\dagger U = I$,

$$g_U(XU, YU) = \text{Re Tr} \left[X^\dagger Y \right].$$

The Riemannian gradient $\text{grad } C(U)$ is the unique tangent vector satisfying

$$dC(U)(\Omega U) = g_U(\text{grad } C(U), \Omega U)$$

for every tangent vector $\Omega U \in T_U \text{SU}(p)$.

The negative-gradient flow is

$$\dot{U}(t) = -\text{grad } C(U(t)).$$

Along this flow,

$$dC(U)(\text{grad } C(U)) = -g_U(\text{grad } C(U), \text{grad } C(U)) \leq 0.$$

Thus the negative Riemannian gradient flow is the manifold analogue of ordinary steepest descent.

2.4 Deriving the Riemannian gradient for a general Hermitian B

We now compute the Riemannian gradient of

$$C(U) = \operatorname{Re} \operatorname{Tr} [HUBU^\dagger].$$

Let

$$U(t) = e^{t\Omega}U, \quad \Omega \in \mathfrak{su}(p).$$

This is a curve through U , and its tangent vector at $t = 0$ is

$$\dot{U}(0) = \Omega U.$$

Computing the differential of the objective. We find

$$dC(U)(\Omega U) = \left. \frac{d}{dt} C(e^{t\Omega}U) \right|_{t=0} \quad (107)$$

$$= \left. \frac{d}{dt} \operatorname{Tr} [HU(t)BU^\dagger(t)] \right|_{t=0} \quad (108)$$

$$= \operatorname{Tr} [H(\dot{U}(t)BU^\dagger(t) + U(t)B\dot{U}^\dagger(t))] \Big|_{t=0} \quad (109)$$

$$= \operatorname{Tr} [H[\Omega, UBU^\dagger]]. \quad (110)$$

Therefore

$$dC(U)(\Omega U) = \operatorname{Tr} [UBU^\dagger, H]\Omega.$$

Hence

$$[UBU^\dagger, H] \in \mathfrak{su}(p).$$

The previous calculation gives the directional derivative in every tangent direction ΩU . To identify the Riemannian gradient, write

$$\operatorname{grad} C(U) = GU, \quad G \in \mathfrak{su}(p).$$

By definition,

$$dC(U)(\Omega U) = g_U(GU, \Omega U)$$

for every $\Omega \in \mathfrak{su}(p)$.

Identifying the Riemannian gradient. Remember that

$$dC(U)(\Omega U) = g_U(\operatorname{grad} C(U), \Omega U) \quad (111)$$

$$= \operatorname{Re} \operatorname{Tr} [(\operatorname{grad} C(U))^\dagger \Omega U]. \quad (112)$$

But then we immediately see that

$$\boxed{\text{grad } C(U) = -[UBU^\dagger, H]U}.$$

For minimization, we follow the negative Riemannian gradient flow:

$$\dot{U} = -\text{grad } C(U).$$

Hence

$$\boxed{\dot{U} = [UBU^\dagger, H]U}.$$

2.5 The induced double-bracket flow for $B(t)$

Deriving $\dot{B}$. The moving matrix is

$$B(t) = U(t)BU^\dagger(t).$$

Differentiating,

$$\dot{B}(t) = \dot{U}(t)BBU^\dagger B(t) + UB(t)B\dot{U}^\dagger B(t) \quad (113)$$

We obtain

$$\boxed{\dot{B} = [[B, H], B]}.$$

This is the Schrödinger-picture double-bracket flow associated with the general Hermitian matrix C .

2.6 The Heisenberg-picture double-bracket flow for $H(t)$

The same unitary flow can be written in the Heisenberg picture. Define

$$H(t) = U(t)^\dagger H U(t).$$

Then

$$\text{Tr}[HC(t)] = \text{Tr}[H(t)B].$$

Thus, instead of evolving $C(t)$ while holding H fixed, we may evolve $H(t)$ while holding B fixed, which gives

$$\boxed{\dot{H} = [H, [B, H]]}.$$

2.7 Discretizing the general double-bracket flow

The continuous-time unitary flow is

$$\dot{U} = [UBU^\dagger, H]U,$$

A first-order (Euler) discretization gives

$$U_{k+1} = U_k + \eta[B_k, H]U_k, \quad \eta > 0.$$

But we have to retract the gradient to the manifold.

$$U_{k+1} = \exp(\eta[B_k, H])U_k, \quad \eta > 0.$$

At iteration k , define

$$B_k = U_k B U_k^\dagger, \quad H_k = U_k^\dagger H U_k.$$

The induced Schrödinger-picture update is

$$B_{k+1} = \exp(\eta[B_k, H]) B_k \exp(-\eta[B_k, H]).$$

The induced Heisenberg-picture update is

$$H_{k+1} = \exp(-\eta[B, H_k]) H_k \exp(\eta[B, H_k]).$$

Key point: You can now implement this in a classical computer.

2.8 Special case I: $B = \rho_0$, the state double-bracket flow

Now choose

$$B = \rho_0, \quad \rho_0 = |\psi_0\rangle\langle\psi_0|.$$

The general Schrödinger-picture flow becomes

$$\dot{\rho} = [[\rho, H], \rho].$$

This is the state double-bracket flow.

$$\rho_{k+1} = \exp(\eta[\rho_k, H]) \rho_k \exp(-\eta[\rho_k, H]).$$

Key point: Converges when ρ_k is an eigenstate of H .

For n qubits, a basis for the Lie algebra can be obtained from Pauli words. Let P_a denote an n -qubit Pauli string. Then iP_a is skew-Hermitian, and we may expand

$$[\rho_k, H] = \sum_a \alpha_a^{(k)} iP_a.$$

The coefficients are determined by projecting the commutator onto Pauli directions. Equivalently, they are related to expectation values of commutators,

$$\alpha_a^{(k)} \propto \text{Tr}[\rho_k i[H, P_a]].$$

The exact update would require all $4^n - 1$ nontrivial Pauli directions:

$$U_{k+1} = \exp \left(\eta \sum_a \alpha_a^{(k)} i P_a \right) U_k.$$

Question: What is the problem with this approach?

Answer: This is generally not efficient, because a generic element of $\mathfrak{su}(2^n)$ contains exponentially many directions.

The exact update

$$U_{k+1} = \exp(\eta[\rho_k, H]) U_k, \quad \rho_k = U_k \rho_0 U_k^\dagger,$$

is not directly useful unless we know how to implement the exponential of the commutator. One way to proceed is to expand the commutator in a Pauli basis.

In this subsection K_a denotes a Hermitian Pauli word, such as $X_1 Z_3 Y_4$. The corresponding Lie-algebra direction is iK_a . Thus

$$K_a^\dagger = K_a, \quad K_a^2 = I, \quad iK_a \in \mathfrak{su}(2^n)$$

for all non-identity Pauli words.

Let

$$\mathfrak{k} = \text{span}\{iK_a : a = 1, \dots, q\} \subseteq \mathfrak{su}(2^n)$$

be the set of directions we are willing to use. This may be the full Pauli basis, or a smaller set such as all one- and two-local Paulis. The projected commutator has the form

$$[\rho_k, H]_{\mathfrak{k}} = \sum_{a=1}^q \alpha_a^{(k)} iK_a.$$

Using the Hilbert–Schmidt inner product and

$$\text{Tr}(K_a K_b) = 2^n \delta_{ab},$$

the coefficients are

$$\alpha_a^{(k)} = -\frac{i}{2^n} \text{Tr}([\rho_k, H] K_a) \tag{114}$$

$$= -\frac{1}{2^n} \text{Tr}(\rho_k i[H, K_a]). \tag{115}$$

This is equivalent to

$$\alpha_a^{(k)} = \frac{i}{2^n} \langle \psi_k | [K_a, e^{itK_a} H e^{-itK_a}] | \psi_k \rangle \Big|_{t=0} \tag{116}$$

So we can use the parameter shift rule to find this coefficient:

$$\alpha_a^{(k)} = \frac{1}{2^n} \langle \psi_k | e^{i\pi/2K_a} H e^{-i\pi/2K_a} | \psi_k \rangle - \langle \psi_k | e^{-i\pi/2K_a} H e^{i\pi/2K_a} | \psi_k \rangle \Big|_{t=0} \tag{117}$$

This perspective connects variational quantum algorithms to dynamical control. The commutator $[\rho, H]$ gives the ideal control direction on the full unitary group, while the restricted subspace $\mathfrak{k}$ encodes the control Hamiltonians that are actually available. The resulting algorithm is a controlled, discretized approximation to the double-bracket gradient flow.

Key point: In practice, this may not converge due to the approximation error.

2.9 Special case II: $B = D$, the diagonal-operator flow

Now choose B to be a diagonal Hermitian matrix:

$$B = D.$$

If D is nondegenerate, then the condition

$$[D, H_k] = 0$$

forces H_k to be diagonal in the computational basis. The iteration scheme becomes

$$H_{k+1} = \exp(-\eta[D, H_k]) H_k \exp(\eta[D, H_k]), \quad H_0 = H$$

Thus the flow drives $H(t)$ toward a matrix that commutes with D . If D is diagonal and nondegenerate, this means that $H(t)$ is driven toward a diagonal matrix.

Why D is useful for DB-DOI. The exact Heisenberg-picture update is generated by the commutator

$$[D, H_k].$$

A first-order step of the descent convention is

$$H_{k+1} = e^{-\eta[D, H_k]} H_k e^{\eta[D, H_k]}, \quad \eta > 0.$$

Directly implementing

$$e^{-\eta[D, H_k]}$$

is generally difficult, because $[D, H_k]$ may be a complicated many-body operator.

DB-DOI replaces this difficult commutator exponential by a group-commutator approximation. For $r = \sqrt{\eta}$,

$$e^{-\eta[D, H_k]} \approx e^{-irH_k} e^{irD} e^{irH_k} e^{-irD} \equiv G_k,$$

up to higher-order errors in η . Since

$$H_k = U_k^\dagger H U_k,$$

we have

$$e^{itH_k} = U_k^\dagger e^{itH} U_k.$$

Therefore the algorithm only needs evolutions under the original Hamiltonian H , evolutions under the simple diagonal operator D , and the circuit U_k already constructed.

$$|\psi\rangle \text{ --- } \boxed{e^{-irH_k}} \text{ --- } \boxed{e^{irD}} \text{ --- } \boxed{e^{irH_k}} \text{ --- } \boxed{e^{-irD}} \text{ --- } \quad (118)$$

Expanded:

$$|\psi_0\rangle \text{---} \boxed{U_k} \text{---} \boxed{U_k^\dagger e^{-irH} U_k} \text{---} \boxed{e^{irD}} \text{---} \boxed{U_k^\dagger e^{irH} U_k} \text{---} \boxed{e^{-irD}} \text{---} \quad (119)$$

This is the central practical advantage of the diagonal-operator choice: instead of measuring all coefficients of the full double-bracket direction, one uses coherent Hamiltonian evolutions and a simple diagonal generator to approximate the flow.

Key point: This circuit will diagonalize it in the ordering chosen by D !

Summary of the two important choices.

Choice of B	Moving object	Flow	Interpretation
ρ_0	$\rho(t) = U(t)\rho_0 U(t)^\dagger$	$\dot{\rho} = [[\rho, H], \rho]$	state-preparation flow
D	$H(t) = U(t)^\dagger H U(t)$	$\dot{H} = [H, [D, H]]$	diagonalization / DB-DOI flow

The choice $B = \rho_0$ gives a flow directly on quantum states and decreases the physical energy

$$\text{Tr}(H\rho).$$

The choice $B = D$ gives a flow that rotates the Hamiltonian toward a diagonal form in the basis selected by D . This is the form most naturally connected to DB-DOI and scalable group-commutator approximations.

Key point: This algorithm has beautiful convergence guarantees. The properties of this optimization scheme are well understood, in particular, if H is non-degenerate there exist exactly $p!$ minima on $SU(p)$, and $(p-1)!$ global minima. Amazingly, only the global minima are stable attractors of the optimization dynamics, and one can show that almost all points will converge to these minima given a suitable step size.

Question: How can we turn this into a quantum algorithm?

Answer: There are two complementary routes. First, we can estimate the projected double-bracket direction by measuring commutator coefficients using the parameter-shift rule. Second, we can avoid estimating all coefficients by using a coherent group-commutator approximation to the double-bracket flow.

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